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# (54) OVERHEAD CONVEYING VEHICLE

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*Assistant Examiner* — Keith R Campbell
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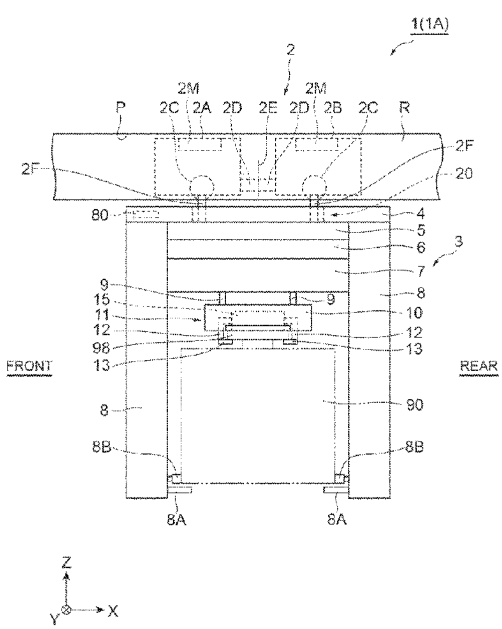
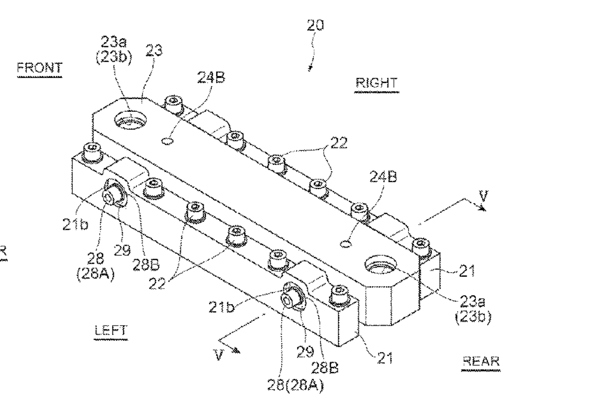
(57) **ABSTRACT**

An overhead conveying vehicle in which a main body part holding a conveyed object is suspended and supported by a traveling part includes a base unit suspended by the traveling part, a horizontal shaft integrally provided with the base unit and extending in a horizontal direction, and a support part provided on the main body part and supporting the horizontal shaft via a rubber bush.

**8 Claims, 11 Drawing Sheets**



(51) **Int. Cl.**
***B61B 13/06*** (2006.01)
***B65G 1/04*** (2006.01)
***B65G 47/61*** (2006.01)

(52) **U.S. Cl.**
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See application file for complete search history.

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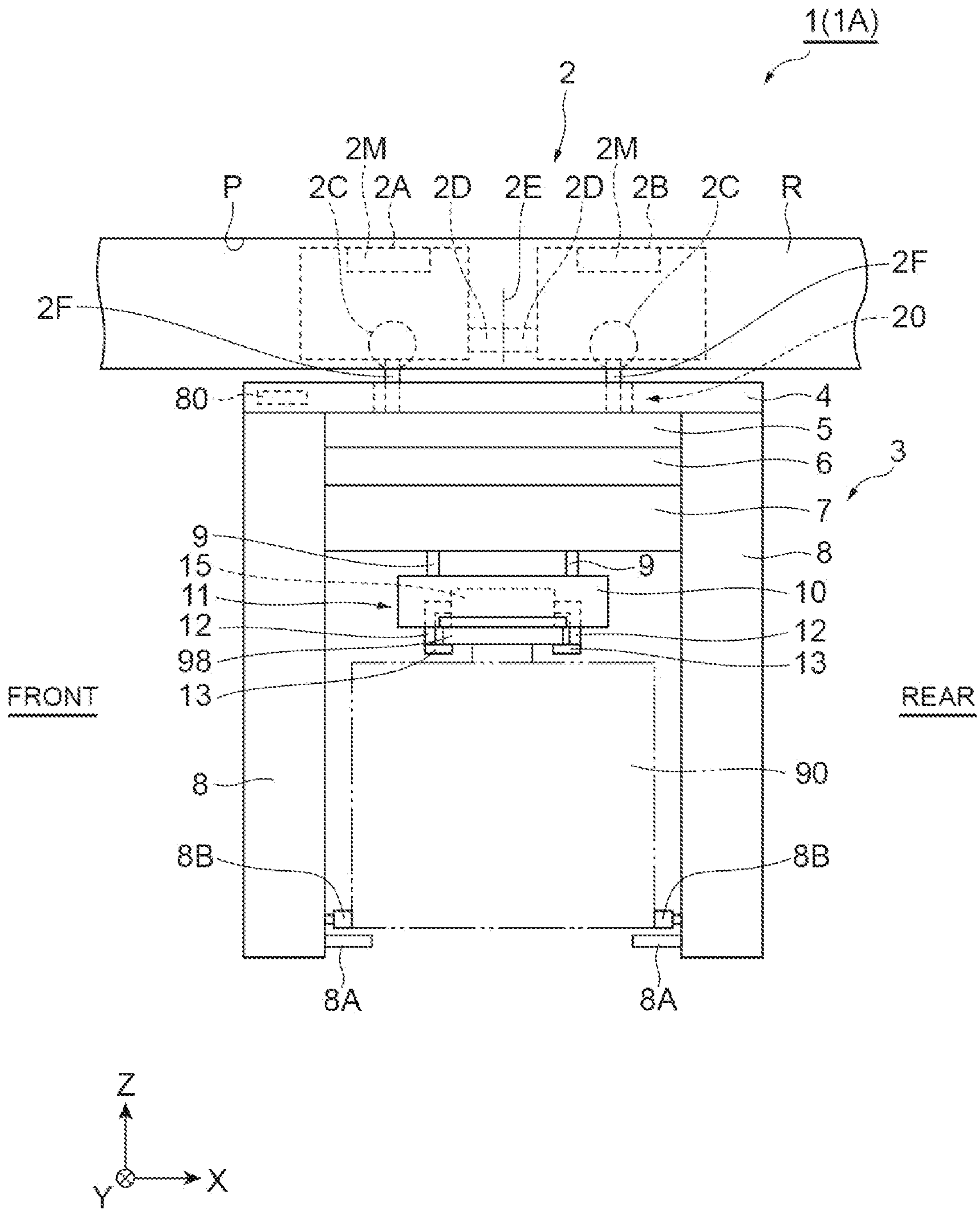
Fig.1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80
4
5
6
3
7
9
9
8
15
10
11
12
12
98
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13
FRONT
REAR
8
90
8B
8B
8A
8A
Z
Y
X

Fig.2

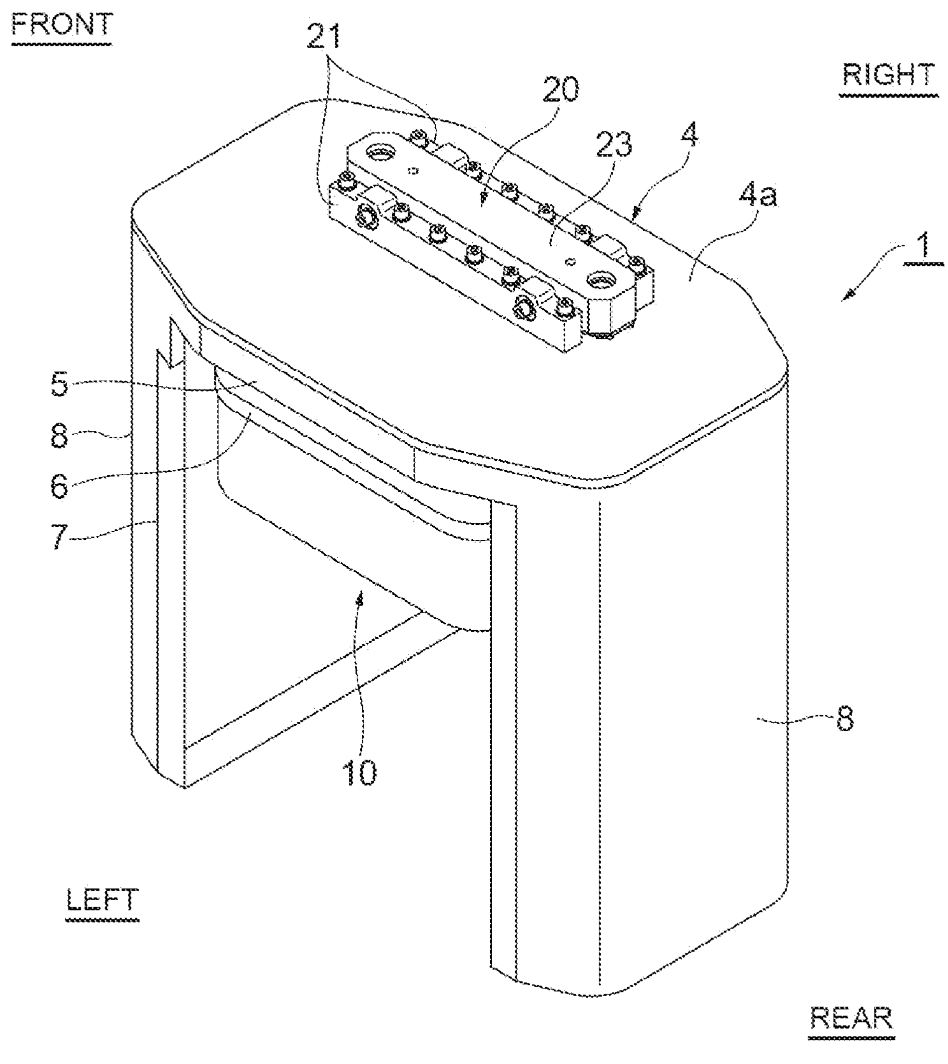
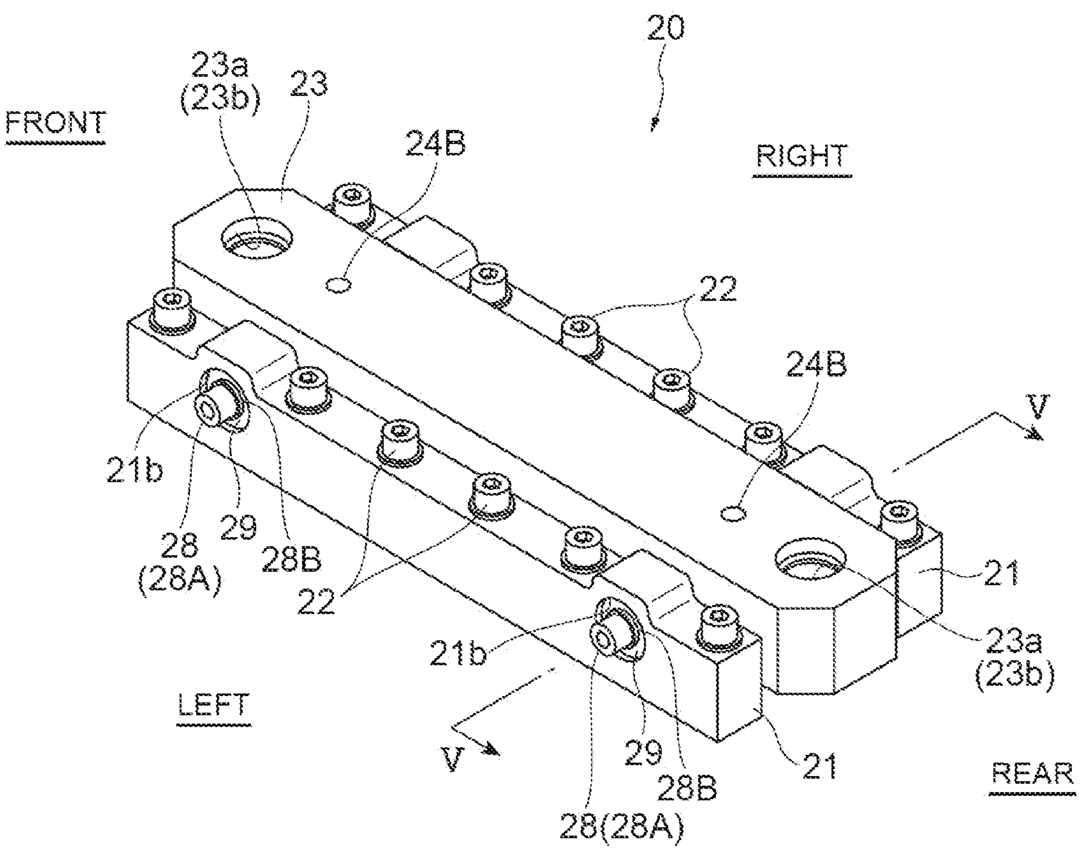


Fig.3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23a
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23
FRONT
24B
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24B
V
21b
28
(28A)
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28B
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28(28A)
21
REAR

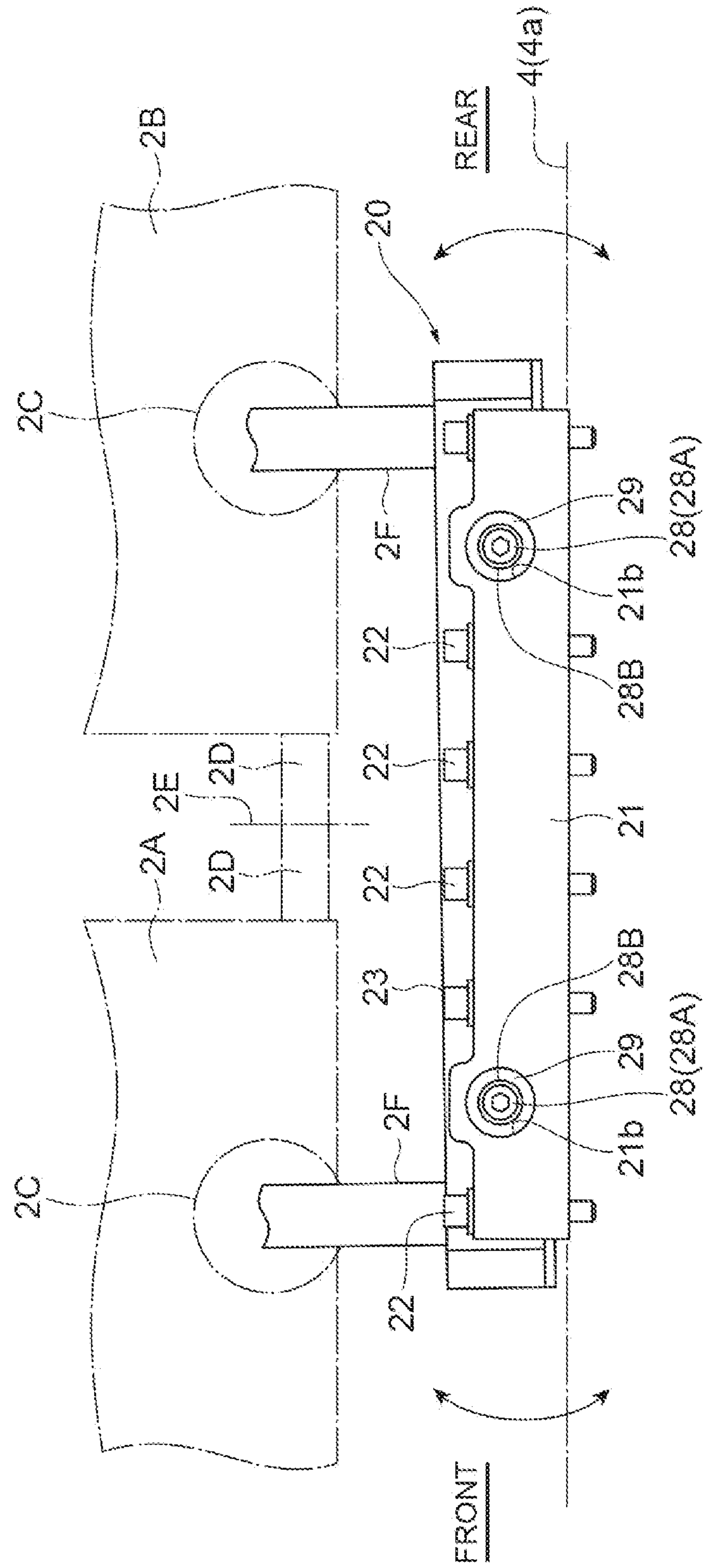
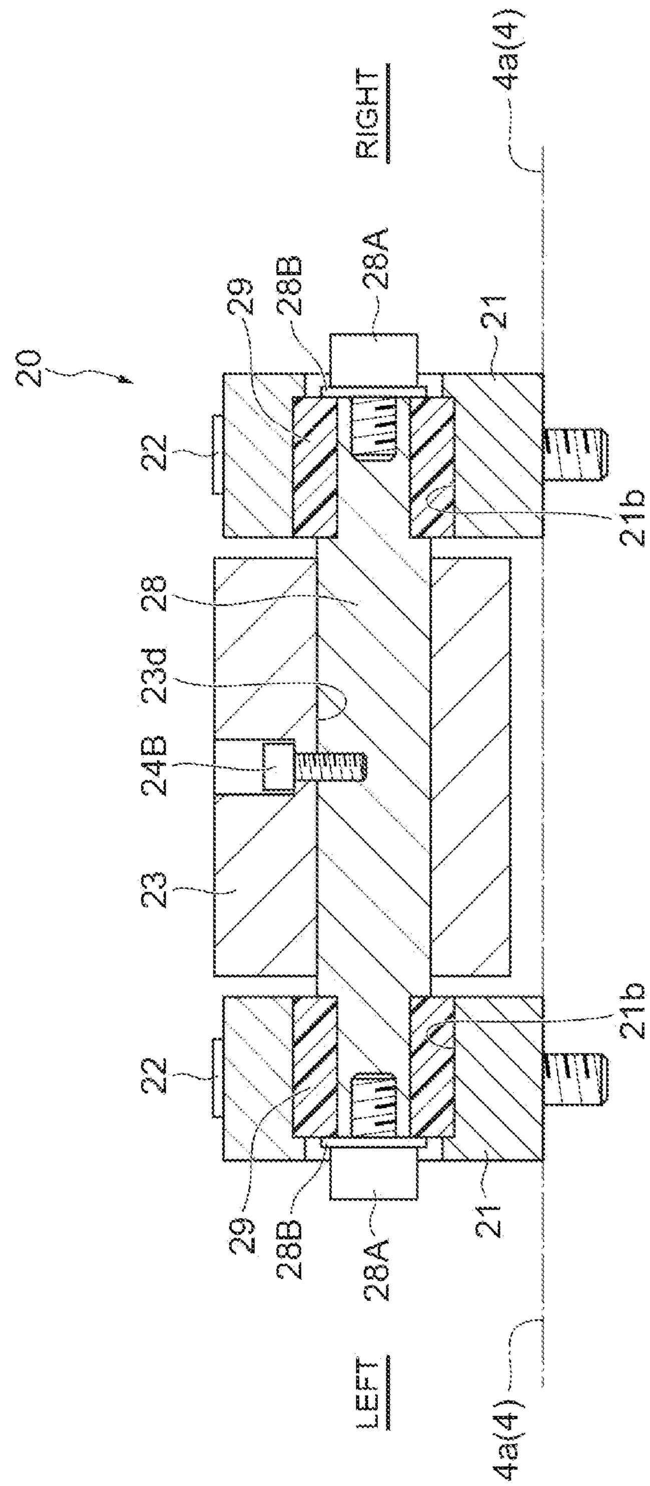



Fig.4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RIGHT
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4a(4)
4a(4)
22
22
29
29
28B
28B
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28A
21
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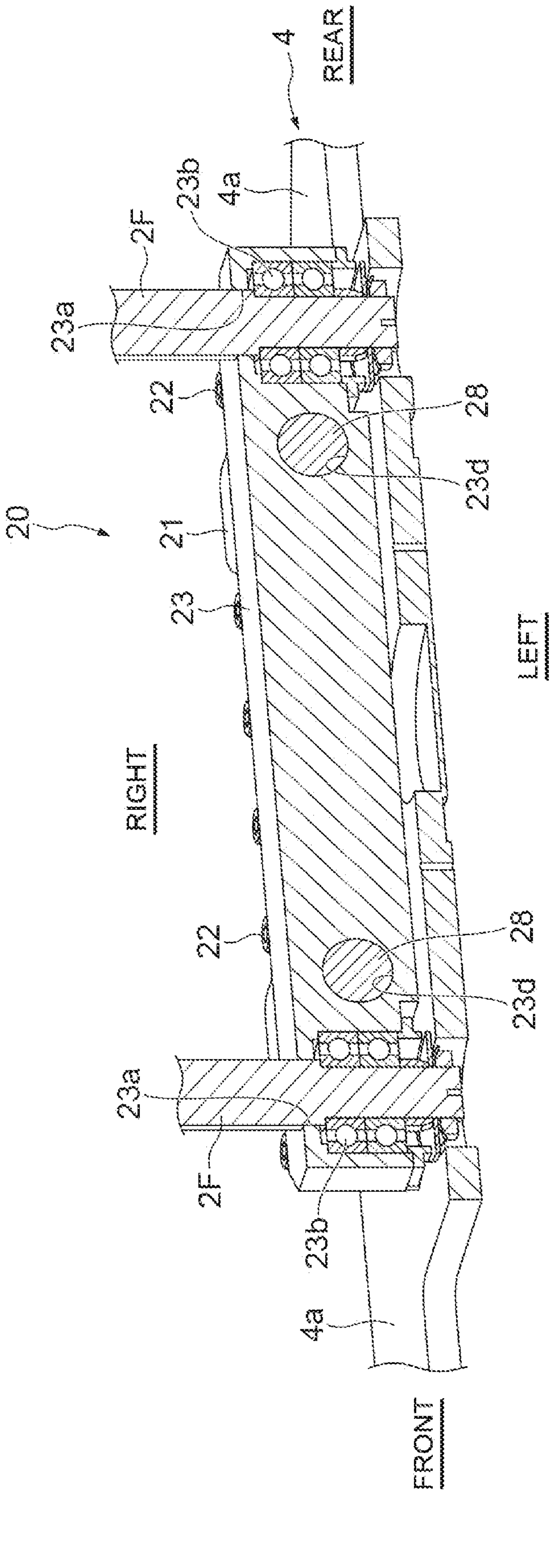
21b
21b
28
23d
24B
23

Fig.5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23b
2F
23a
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23d
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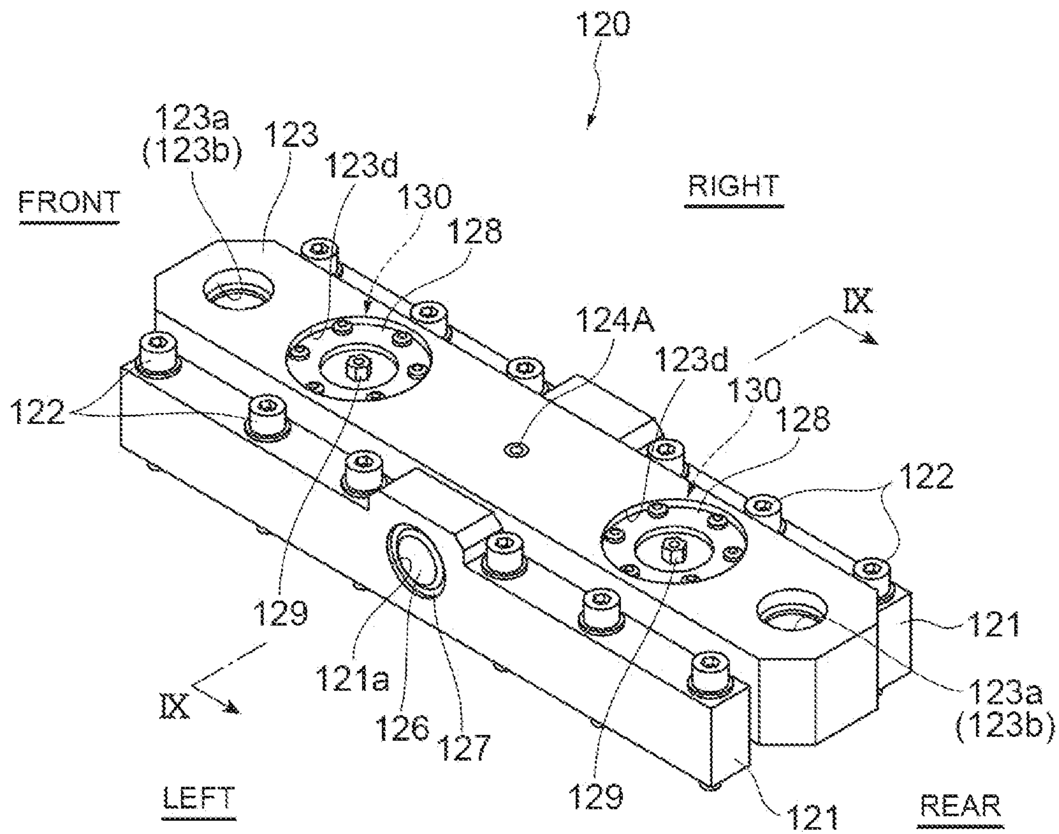
23b
4a
FRONT

Fig.6

Fig.7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123d
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128
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124A
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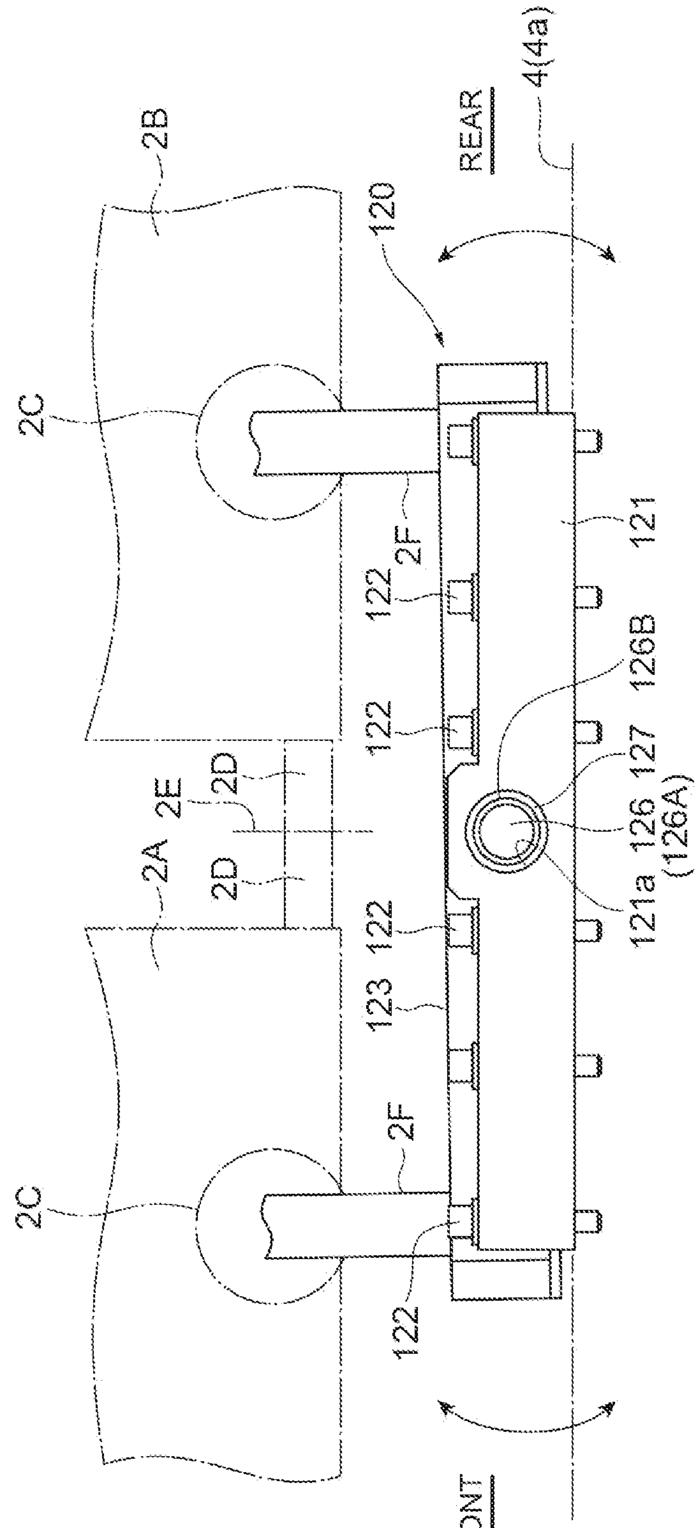
121a
126
127
121
123a
(123b)
IX
LEFT
129
121
REAR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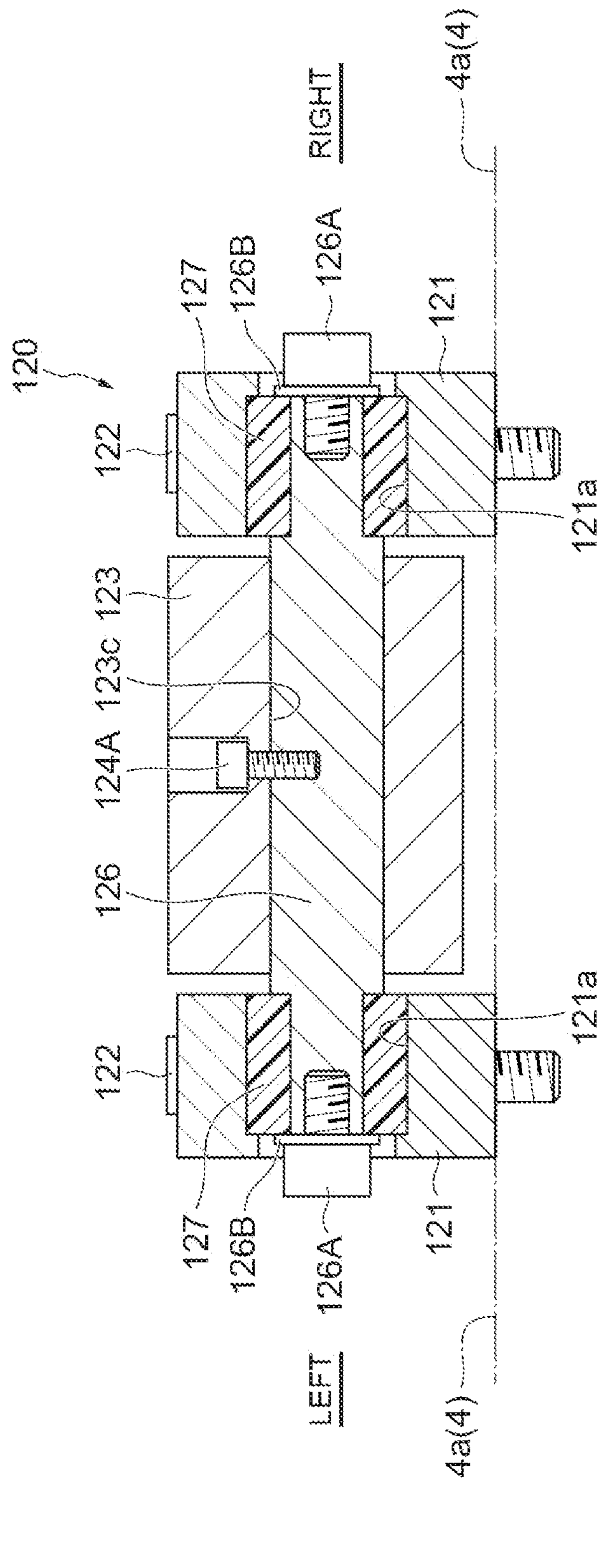
2C
2F
121
122
126B
127
126
(126A)
121a
2D
2E
2D
2A
123
122
2F
2C
122
FRONT

Fig.8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RIGHT
LEFT
4a(4)
4a(4)
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126B
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121
121a
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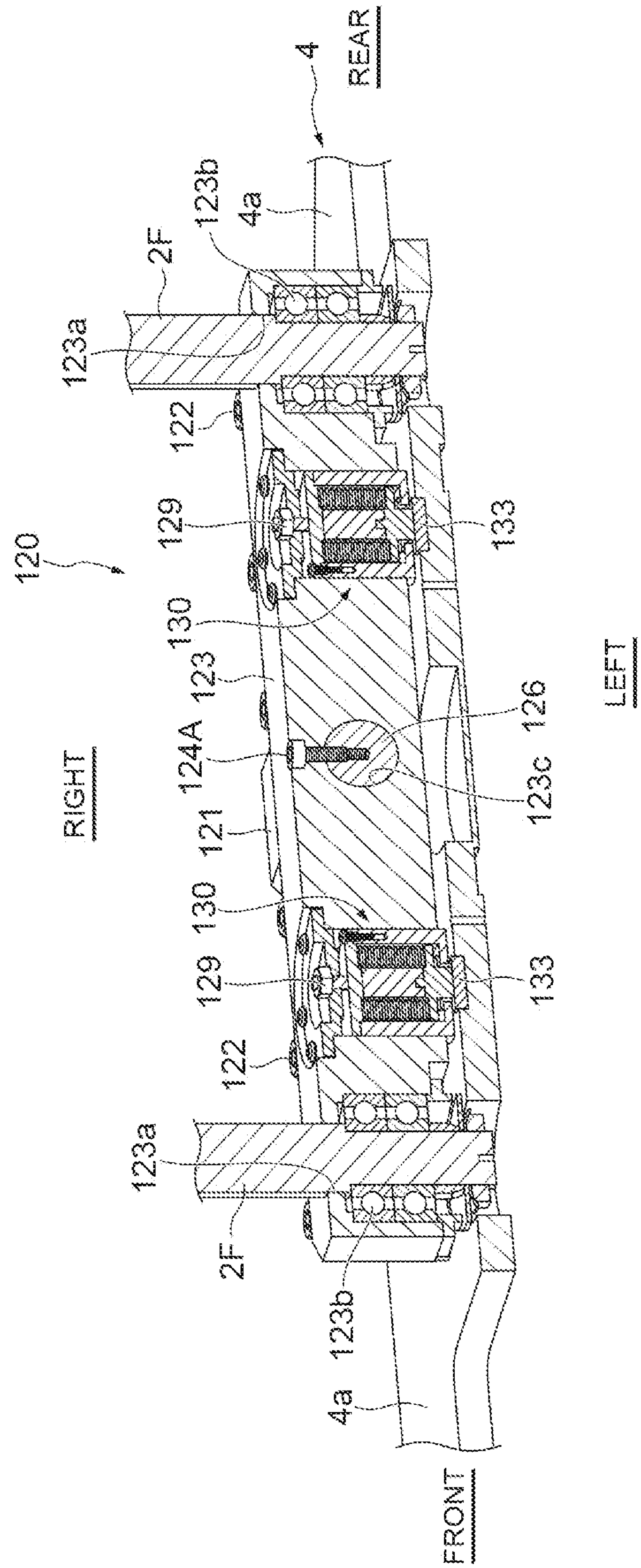
123c
124A
126
122
121a
127
126B
126A
121

Fig.9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123b
2F
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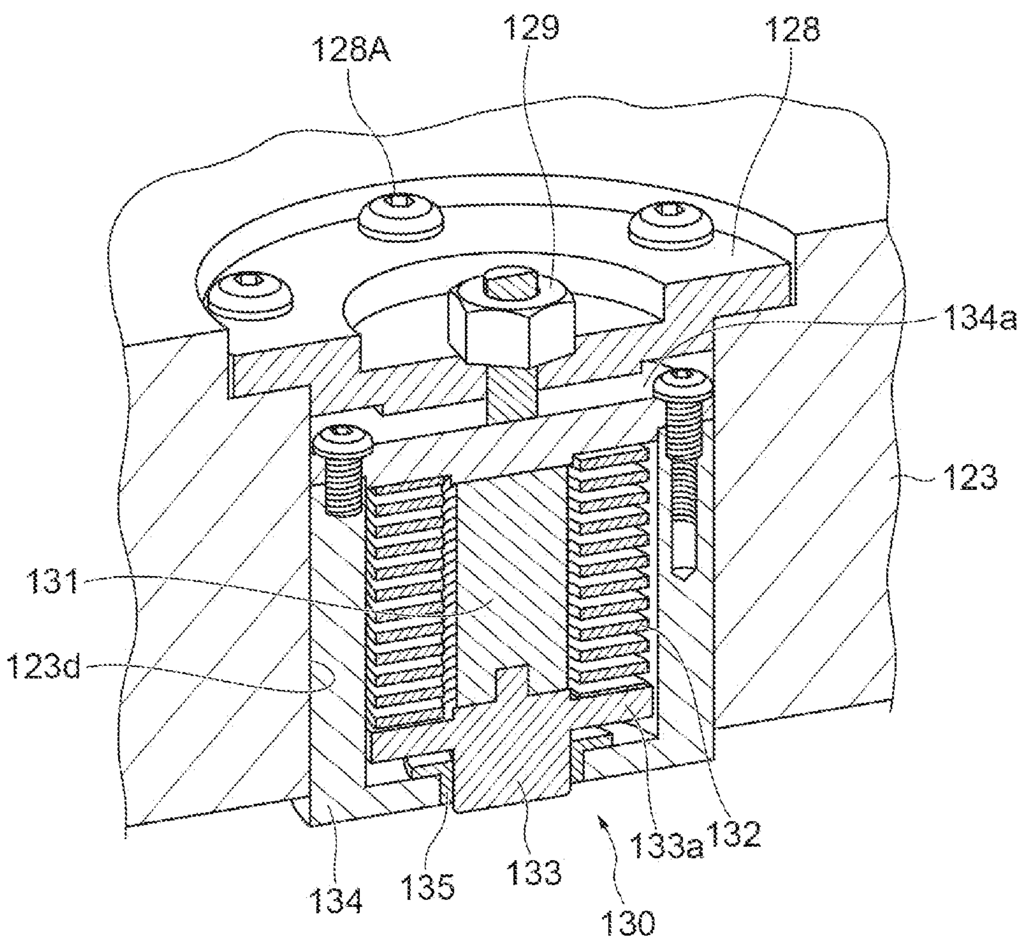
133
120
130
123
LEFT
126
124A
123c
RIGHT
121
130
129
133
122
123a
2F
123b
4a
FRONT

Fig.10

*Fig.11*



# OVERHEAD CONVEYING VEHICLE

## TECHNICAL FIELD

This disclosure relates to an overhead conveying vehicle.

## BACKGROUND

Japanese Unexamined Patent Publication No. 2001-298065 discloses an automated guided vehicle (AGV) including a cart, an AGV body placed on the cart via a damper for vibration isolation, and a transfer device provided to the AGV body. The AGV of Japanese Unexamined Patent Publication No. 2001-298065 can reduce vibrations transmitted to an article while traveling.

However, an overhead conveying vehicle with a structure in which the main body part holding an article (conveyed object) is suspended and supported by a traveling part cannot be configured to have the main body part placed on top of the traveling part via a damper for vibration isolation, as shown in Japanese Unexamined Patent Publication No. 2001-298065.

Therefore, it could be helpful to provide an overhead conveying vehicle capable of suppressing propagation of vibration from the traveling part to the conveyed object, even if the overhead conveying vehicle has a configuration in which the main body part holding the conveyed object is suspended and supported by the traveling part.

## SUMMARY

We thus provide:

An overhead conveying vehicle is an overhead conveying vehicle in which a main body part holding a conveyed object is suspended and supported by a traveling part, and includes a base unit suspended by the traveling part, a horizontal shaft integrally provided with the base unit and extending in a horizontal direction, and a support part provided with the main body part and supporting the horizontal shaft via a rubber bush.

The horizontal shaft integrally configured in the base unit, which is suspended from the traveling part, is supported by the support part via the rubber bush. In other words, the traveling part and the main body part are connected via the rubber bush. With this configuration, vibration transmitted from the traveling part to the base unit is absorbed by the rubber bush. As a result, even an overhead conveying vehicle with a configuration in which a main body part holding a conveyed object is suspended and supported by a traveling part can suppress propagation of vibration from the traveling part to the conveyed object.

The horizontal shaft may be disposed to extend in a direction orthogonal to both a traveling direction of the traveling part and a vertical direction. In this configuration, when an angular change of the traveling part is allowed with respect to the main body part, the angular change of the traveling part is allowed only in the traveling direction of the traveling part so that tilting of the main body part can be minimized when the conveyed object is transferred in a right-and-left direction (lateral direction) perpendicular to the traveling direction.

A plurality of the horizontal shafts may be provided. In this configuration, a relative positional relationship between the traveling part and the main body part (posture of the main body part with respect to the traveling part) can be maintained more stably.

Only one piece of the horizontal shaft may be provided, and the overhead conveying vehicle may further include a pair of dampers provided in the base unit in a manner of sandwiching the horizontal shaft in the traveling direction of the traveling part and configured to contact the main body part. In this configuration, not only the propagation of vibration can be suppressed but also the vibration can be damped in an early stage. Furthermore, in this configuration, the relative positional relationship between the main body part and the traveling part can be maintained more stably.

The base unit may be suspended by the traveling part via a pair of suspension members extending downward in the vertical direction from the traveling part, and the base unit may rotatably support each of the suspension members around an axis as a turning axis that extends in an extending direction of the suspension members. With this configuration, in the suspension member, it is possible to reduce occurrence of torsional stress between an attachment portion to the traveling part and an attachment portion to the base unit.

An overhead conveying vehicle with a configuration in which a main body part holding a conveyed object is suspended and supported by a traveling part can suppress propagation of vibration from the traveling part to the conveyed object.

## BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a side view of an overhead conveying vehicle according to a first example when viewed from side.

FIG. **2** is a perspective view of a main body frame of the overhead conveying vehicle of FIG. **1** when viewed from obliquely above.

FIG. **3** is a perspective view of a vibration isolation unit when viewed from obliquely above.

FIG. **4** is a side view of the vibration isolation unit when viewed from side.

FIG. **5** is a schematic plan view when viewed from a line V-V of FIG. **3**.

FIG. **6** is a sectional view of the vibration isolation unit cut along a front-and-rear direction.

FIG. **7** is a perspective view of a vibration isolation unit according to a second example when viewed from obliquely above.

FIG. **8** is a side view of the vibration isolation unit according to the second example when viewed from side.

FIG. **9** is a schematic plan view when viewed from a line IX-IX of FIG. **7**.

FIG. **10** is a sectional view of the vibration isolation unit according to the second example cut along a front-and-rear direction.

FIG. **11** is a sectional perspective view illustrating an inner structure of a damper of FIG. **10**.

## REFERENCE SIGNS LIST

**1**, **1**A overhead conveying vehicle
**2** traveling part
**2**F suspension member
**3** main body part
**4** main body frame
**20** vibration isolation unit
**21** support part
**23** base unit
**28** horizontal shaft
**29** rubber bush
**90** conveyed object

120 vibration isolation unit
121 support part
123 base unit
126 horizontal shaft
127 rubber bush
130 damper
131 rubber member
132 spring member
133 contact part
R traveling rail

## DETAILED DESCRIPTION

An overhead conveying vehicle **1** according to an example will now be described with reference to the drawings. In the description of the drawings, like elements are designated by like reference signs, and duplicate description is omitted.

### First Example

An overhead conveying vehicle **1** according to a First Example will now be described. The overhead conveying vehicle **1** illustrated in FIG. **1** travels along a traveling rail R installed at a position higher than a floor such as on a ceiling of a clean room. The overhead conveying vehicle **1** transports a conveyed object **90** between a storage facility and a predetermined load port, for example. Examples of the conveyed object **90** include containers such as a front opening unified pod (FOUP) that store a plurality of semiconductor wafers and reticle pods that store glass substrates, and general components and the like. The containers described above have a flange **98** to be held by the overhead conveying vehicle **1**.

In the following description, for convenience of explanation, the right-and-left direction (X-axis direction) in FIG. **1** is defined as "front-and-rear direction" (second direction) of the overhead conveying vehicle **1**. The up-and-down direction (Z-axis direction) in FIG. **1** is defined as an up-and-down direction (vertical direction) of the overhead conveying vehicle **1**. A depth direction (Y-axis direction) in FIG. **1** is defined as a right-and-left direction or a width direction (first direction) of the overhead conveying vehicle **1**. The X-axis, the Y-axis, and the Z-axis are orthogonal to each other.

As illustrated in FIGS. **1** and **2**, the overhead conveying vehicle **1** includes a traveling part **2**, a main body part **3**, and a lifting part **10**. The traveling part **2** moves the overhead conveying vehicle **1** along the traveling rail R. The traveling part **2** is disposed inside the traveling rail R.

The traveling part **2** has a front traveling body **2**A and a rear traveling body **2**B. A connection part **2**D provided on the front traveling body **2**A and a connection part **2**D provided on the rear traveling body **2**B are rotatably connected by a connection shaft **2**E. The front traveling body **2**A and the rear traveling body **2**B are provided with respective traveling rollers **2**C, **2**C and respective traveling drive parts **2**M, **2**M. The traveling drive part **2**M is an LDM (Linear DC Motor) that accelerates or brakes the overhead conveying vehicle **1** by magnetic force generated between the LDM and a magnetic plate P disposed on a top surface of the traveling rail R.

The main body part **3** is suspended and supported by the traveling part **2**. More precisely, the main body part **3** is suspended and supported by the traveling part **2** when a pair of suspension members **2**F, **2**F extending downward in the vertical direction from the traveling part **2** are connected to a vibration isolation unit **20** provided in a main frame **4** of the main body part **3**. Details of the vibration isolation unit **20** are described in detail in a later section. The main body part **3** has a main frame **4**, a horizontal drive part **5**, a rotary drive part **6**, a lifting drive part **7**, a lifting part **10**, a pair of covers **8**, **8**, and a controller **80**.

The horizontal drive part **5** is fixed to a lower part of the main frame **4**. The horizontal drive part **5** moves the rotary drive part **6**, the lifting drive part **7**, and the lifting part **10** in a direction orthogonal to the extending direction of the traveling rail R (right-and-left direction) in a horizontal plane. The rotary drive part **6** rotates the lifting drive part **7** and the lifting part **10** in the horizontal plane. The lifting drive part **7** raises and lowers the lifting part **10** by winding and paying out four belts **9**. As the belts **9** in the lifting drive part **7**, appropriate suspending members such as wires or ropes may be used.

The lifting part **10** is provided to be able to be raised and lowered by the lifting drive part **7**, and functions as a lifting platform in the overhead conveying vehicle **1**. The lifting part **10** includes a holding device **11** configured to grip the conveyed object **90**, and is raised and lowered by the belts **9** with respect to the horizontal drive part **5**, the rotary drive part **6**, and the lifting drive part **7** included in the main body part. The holding device **11** holds the conveyed object **90**. The holding device **11** includes a pair of arms **12**, **12** formed in a L shape, hook parts **13**, **13** fixed to the respective arms **12**, **12**, and an opening/closing mechanism **15** configured to open and close the pair of arms **12**, **12**.

The opening/closing mechanism **15** moves the pair of arms **12**, **12** in a direction coming closer to each other and in a direction separating apart from each other. The pair of arms **12**, **12** are moved forward and backward in the front-and-rear direction in accordance with an operation of the opening/closing mechanism **15**. By this movement, the pair of hook parts **13**, **13** fixed to the arms **12**, **12** are opened and closed.

In the First Example, a height position of the holding device **11** (the lifting part **10**) is adjusted (lowered) so that the holding surface of each hook part **13** is positioned lower than the height of the lower surface of the flange **98** when the pair of hook parts **13**, **13** are in an open state. In this state, when the pair of hook parts **13**, **13** are brought into a closed state, the holding surfaces of the hook parts **13**, **13** are moved forward below the lower surface of the flange **98**, and the lifting part **10** is raised in this state so that the flange **98** is held (gripped) by the pair of hook parts **13**, **13**, and the conveyed object **90** is supported.

The pair of covers **8**, **8** are provided on the front and the rear of the traveling direction to cover the horizontal drive part **5**, the rotary drive part **6**, the lifting drive part **7**, the lifting part **10**, and the holding device **11**. The pair of covers **8**, **8** form a space for accommodating the conveyed object **90** below the holding device **11** in a state in which the lifting part **10** has ascended to its ascending end. Each of the paired covers **8**, **8** has a drop prevention mechanism **8**A and a swaying suppression mechanism **8**B. The drop prevention mechanism **8**A prevents the conveyed object **90** held by the holding device **11** from falling in the state in which the lifting part **10** has ascended to its ascending end. The swaying suppression mechanism **8**B suppresses swaying of the conveyed object **90** held by the holding device **11** in the front-and-rear direction (traveling direction) and the right-and-left direction of the overhead conveying vehicle **1** during traveling.

The controller **80** is an electronic control unit including a central processing unit (CPU), a read only memory (ROM), a random access memory (RAM) and the like. The controller **80** controls various operations of the overhead conveying vehicle **1**. Specifically, the controller **80** controls the traveling part **2**, the horizontal drive part **5**, the rotary drive part **6**, and the lifting drive part **7**. The controller **80** can be configured as software, for example, in which a program stored in the ROM is loaded onto the RAM and executed by the CPU. The controller **80** may be configured as hardware including electronic circuitry or the like. The controller **80** communicates with a higher-level controller (not illustrated) using a power feeding part (power feeding line) or a feeder line on the traveling rail R.

The following describes a vibration isolation unit **20** with reference to FIGS. **3** to **6**. The vibration isolation unit **20** includes a pair of support parts **21**, **21**, a base unit **23**, and a pair of horizontal shafts **28**, **28**.

The pair of support parts **21**, **21** are arranged spaced apart in the right-and-left direction and are fixed to a top surface **4***a* of the main frame **4**. The pair of support parts **21**, **21** are fixed to the main frame **4** by members such as screws **22**, for example. The pair of horizontal shafts **28**, **28** extending horizontally in the right- and left direction are crossed over the pair of support parts **21**, **21**. More precisely, the pair of support parts **21**, **21** respectively have insertion holes **21***b*, **21***b* formed therein through which the pair of horizontal shafts **28**, **28** are respectively inserted. The insertion holes **21***b*, **21***b* are formed at both ends of the base unit **23** in the front-and-rear direction. A rubber bush **29** of a cylindrical shape is interposed in the insertion hole **21***b*, and the horizontal shaft **28** is interposed in the rubber bush **29**. In other words, the horizontal shaft **28** is supported by the pair of support parts **21** via the rubber bush **29**. The rubber bush **29** is formed by a material such as nitrile rubber, for example. At both ends of the horizontal shaft **28** in the extension direction, there are bolts **28**A and washers **28**B that act as detents from the insertion holes **21***b*.

The rubber bushes **29**, **29**, which are respectively interposed in the pair of support parts **21**, **21** may be formed by materials of different hardness from each other. For example, the left and the right rubber bushes **29**, **29** that are made different in hardness from each other can easily maintain the relative position (posture) of a main body part **3** with respect to the traveling part **2** when a position of the center of gravity of the vibration isolation unit **20** and that of the main body part **3** viewed from the Z-axis direction differ from each other, or when the conveyed object **90** is transferred in the right-and-left direction (lateral direction).

The base unit **23** is suspended from the traveling part **2** via the pair of suspension members **2**F, **2**F. The base unit **23** is a member extending in the front-and-rear direction. The base unit **23** has insertion holes **23***a*, **23***a* formed therein through which the pair of suspension members **2**F, **2**F can be inserted, respectively. The insertion holes **23***a*, **23***a* are formed at both ends of the base unit **23** in the front-and-rear direction. Moreover, an inner circumferential surface of the insertion hole **23***a* is provided with a bearing **23***b* that rotatably supports the suspension member **2**F around an axis along the extending direction of the suspension member **2**F.

The base unit **23** is provided to not be rotatable with respect to the turning shaft **26**. More precisely, the turning shaft **26** is inserted into an insertion hole **23***d* formed in the base unit **23**, and the turning shaft **26** and the base unit **23** are fixed by a screw **24**B. Insertion holes **23***d*, **23***d* are provided at both ends of the base unit **23** in the front-and-rear direction. In other words, the two horizontal shafts **28**, **28** are provided on the base unit **23** in an unrotatable manner. The base unit **23** is supported by the pair of support parts **21**, **21** via the horizontal shafts **28**, **28**.

The following describes a working effect of the overhead conveying vehicle **1** described above. In the overhead conveying vehicle **1** described above, the horizontal shaft **28** that is integrally configured in the base unit **23** suspended from the traveling part **2** is supported by the support part **21** via the rubber bush **29**. In other words, the traveling part **2** and the main body part **3** are connected via the rubber bush **29**. As a result, the vibration transmitted from the traveling part **2** to the base unit **23** is absorbed by the rubber bush **29**. As a result, even if the overhead conveying vehicle **1** has a configuration in which the main body part **3** holding the conveyed object **90** is suspended and supported by the traveling part **2**, vibration from the traveling part **2** can be suppressed from propagating to the conveyed object **90**.

In the overhead conveying vehicle **1** described above, the horizontal shaft **28** is disposed to extend in a direction orthogonal to both the traveling direction and the vertical direction of the traveling part **2** (right-and-left direction). With this configuration, when an angular change of the traveling part **2** is allowed with respect to the main body part **3**, the angular change of the traveling part is allowed only in the traveling direction of the traveling part **2** so that tilting of the main body part **3** can be minimized when the conveyed object **90** is transferred in a right-and-left direction (lateral direction) perpendicular to the traveling direction.

In the overhead conveying vehicle **1** described above, since two horizontal shafts are provided, the relative positional relationship between the main body part **3** and the traveling part **2** (posture of the main body part **3** with respect to the traveling part **2**) can be maintained more stably.

In the overhead conveying vehicle **1** described above, the base unit **23** rotatably supports each of suspension members **2**F, **2**F around an axis as a turning axis that extend in an extending direction of the suspension members **2**F, **2**F, thereby reducing the occurrence of torsional stress between the attachment portion to the traveling part **2** and the attachment portion to the base unit **23** in the suspension members **2**F, **2**F.

## Second Example

An overhead conveying vehicle **1**A according to a Second Example will now be described mainly with reference to the FIG. **1**, and FIGS. **7** to **11**. In the overhead conveying vehicle **1**A according to the Second Example, the configuration of a vibration isolation unit **120** differs from that of the vibration isolation unit **20** of the overhead transfer vehicle **1** of the First Example. The vibration isolation units **120** the configurations of which differ from each other are described in detail, and description of the other configurations is omitted. The vibration isolation unit **120** includes a pair of support parts **121**, **121**, a base unit **123**, and a pair of dampers **130**, **130**.

The pair of support parts **121**, **121** are arranged spaced apart in the right-and-left direction and are fixed to a top surface **4***a* of the main frame **4**. The pair of support parts **121**, **121** are fixed to the main frame **4** by members such as screws **122**, for example. A turning shaft **126** extending horizontally in the right- and left direction is crossed over the pair of support parts **121**, **121**. More precisely, each of the paired support parts **121**, **121** has an insertion hole **121***a* formed therein through which the turning shaft **126** is inserted. A horizontal shaft **126** is attached to the support parts **121**, **121** via bolts **126**A and washers **126**B provided at both ends of the horizontal shaft **126**. A resin bush **127** of a cylindrical shape is interpolated in the insertion hole **121***a*, and the turning shaft **126** is interposed in the resin bush **127**. In other words, the turning shaft **126** is supported by the pair of support parts **121**, **121** with the resin bushes **127**. The rubber bush **127** is formed by a material such as nitrile rubber, for example. The horizontal shaft **126** is disposed in substantially the center of a main body frame **4** in the front-and-rear direction.

The base unit **123** is suspended from the traveling part **2** via the pair of suspension members **2**F, **2**F. The base unit **123** is a member extending in the front-and-rear direction. The base unit **123** has insertion holes **123***a*, **123***a* formed therein through which the pair of suspension members **2**F, **2**F can be inserted, respectively. The insertion holes **123***a*, **123***a* are formed at both ends of the base unit **123** in the front-and-rear direction. Moreover, an inner circumferential surface of the insertion hole **123***a* is provided with a bearing **123***b* that rotatably supports the suspension member **2**F around an axis along the extending direction of the suspension member **2**F.

The base unit **123** is provided to not be rotatable with respect to the turning shaft **126**. More precisely, the turning shaft **126** is inserted into an insertion hole **123***c* formed in the base unit **123**, and the turning shaft **126** and the base unit **123** are fixed by a screw **124**A. The base unit **123** is supported by the pair of support parts **121**, **121** in a rotatable state via the turning shaft **126** extending in the horizontal direction. In other words, the pair of support parts **121**, **121** rotatably support the turning shaft **126** that rotates integrally with the base unit **123**.

The pair of dampers **130**, **130** are provided to sandwich the turning shaft **126** in the front-and-rear direction and to come into contact with the top surface **4***a* of the main frame **4**. The damper **130** is interposed in an insertion hole **123***d* formed in the base unit **123**, and is provided such that part of the damper **130** protrudes downward (to the top surface **4***a* of the main frame **4**). The damper **130** absorbs energy generated when coming into contact with the top surface **4***a* of the main frame **4**.

The damper **130** has a rubber member **131**, a spring member **132**, a contact part **133**, a housing **134**, and a buffer **135**. The rubber member **131** has elasticity, and is formed of, for example, urethane rubber or other materials. The spring member **132** is disposed to have the rubber member **131** interposed therein. The elasticity of the spring member **132** may be greater than that of the rubber member **131**. This allows the elasticity of the rubber member **131** to be added to the contact part **133** when the spring member **132** shrinks to some degree. The contact part **133** is a member that comes into contact with the top surface **4***a* of the main frame **4**. A portion of the lower part of the contact part **133**, the portion coming into contact with the top surface **4***a* of the main frame **4**, is formed of a material having compression-resistance such as ultra-high molecular weight polyethylene. The upper part of the contact part **133** has a flange **133***a* formed therein that comes into contact with a lower end of the rubber member **131** and a lower end of the spring member **132**. The spring member **132** of the Second Example comes into contact with the contact part **133** in a state of being compressed, that is, pressurization is applied thereto. The housing **134** accommodates the rubber member **131**, the spring member **132**, and part of the contact part **133**. The buffer **135** is a resin bush provided to prevent wear due to vertical movement of the contact part **133**.

The base unit **123** is provided with a lid **128** that covers an upper opening of the insertion hole **123***d*. The lid **128** is fixed to the base unit **123** by a screw **128**A or other means. The center of the lid **128** is provided with an adjustment screw **129** to adjust an amount of push by the damper **130** onto the main frame **4** (the amount of protrusion of the contact part **133** from the base unit **123**). A tip of the adjustment screw **129** is in contact with a top surface **134***a* of the housing **134**. The amount of protrusion of the contact part **133** protruding from the lower surface of the base unit **123** can be adjusted by rotating the adjustment screw **129** with respect to the lid **128**.

The following describes a working effect of the overhead conveying vehicle **1**A according to the Second Example described above. In the overhead conveying vehicle **1**A described above, the horizontal shaft **126** integrally configured in the base unit **123**, which is suspended from the traveling part **2**, is supported by the support part **121** via the rubber bush **127**. In other words, the traveling part **2** and the main body part **3** are connected via the rubber bush **127**. As a result, the vibration transmitted from the traveling part **2** to the base unit **123** is absorbed by the rubber bush **127**. As a result, similar to the overhead conveying vehicle **1**A according to the First Example described above, even if the overhead conveying vehicle **1**A has a configuration in which the main body part **3** holding the conveyed object **90** is suspended and supported by the traveling part **2**, the vibration from the traveling part **2** can be suppressed from propagating to the conveyed object **90**.

The horizontal shaft **126** is disposed to extend in a direction orthogonal to both the traveling direction and the vertical direction of the traveling part **2** (right-and-left direction). With this configuration, when an angular change of the traveling part **2** is allowed with respect to the main body part **3**, the angular change of the traveling part is allowed only in the traveling direction of the traveling part **2** so that tilting of the main body part **3** can be minimized when the conveyed object **90** is transferred in a right-and-left direction (lateral direction) perpendicular to the traveling direction.

In the overhead conveying vehicle **1**A according to the Second Example described above, only one piece of the horizontal shaft **126** is provided. In the traveling direction of the traveling part **2**, the pair of dampers **130**, **130** are provided on the base unit **123** in a manner of sandwiching the horizontal shaft **126** and contacting the main body part **3**. In this configuration, not only the propagation of vibration generated in the traveling part **2** to the main body part **3** can be suppressed, but also the vibration can be damped in an early stage. Furthermore, the relative positional relationship between the traveling part **2** and the main body part **3** (posture of the main body part **3** with respect to the traveling part **2**) can be maintained more stably.

Since the damper **130** has the rubber member **131** and a spring member **132**, as illustrated in FIG. **11**, pressurization generated in the damper **130** can be easily adjusted by compressing the spring member **132**. In the damper **130** of the second embodiment, a predetermined pressurization is applied. When both the traveling part **2** and the main body part **3** have no tilt, the contact part **133** of the damper **130** and the top surface **4***a* of the main frame **4** are in contact with each other in a state in which the contact part **133** does not exert any force on the main frame **4**. The contact part **133** can be brought into contact with the top surface **4***a* of the main frame **4** by rotating the adjustment screw **129** to adjust the amount of protrusion of the contact part **133**.

When the traveling part **2** accelerates forward, the base unit **123** in a freely rotatable state is subjected to force in a manner to sink frontward. However, in the damper **130** to which pressurization is applied, even if such force that causes the base unit **123** to sink frontward is applied, the spring member **132** and the rubber member **131** are not compressed until force exceeding a predetermined value (i.e., force exceeding the force applied to the spring member **132** as pressurization) is exerted, and the base unit **123** is not tilted against the main frame **4**. In the damper **130** of the above Second Example, pressurization that can withstand the force generated by an assumed acceleration is set. With this configuration, occurrence of posture change of the main frame **4** can be suppressed during acceleration of the overhead conveying vehicle **1**A. In other words, occurrence of swaying in the conveyed object **90** can be suppressed during acceleration of the overhead conveying vehicle **1**A.

Since pressurization is applied to the dampers **130**, **130** in both the front and the rear directions, the occurrence of the posture change of the main frame **4** can be suppressed also during deceleration of the overhead conveying vehicle **1**A. In other words, since pressurization is set to withstand the force generated by assumed deceleration, occurrence of swaying of the conveyed object **90** can be suppressed during deceleration of the overhead conveying vehicle **1**A.

The First Example and Second Example have been described above, but this disclosure is not limited to these examples. Various modifications can be made without departing from the appended claims.

In the First Example described above, a configuration is described in which the two horizontal shafts **28**, **28** are provided, and in the Second Example described above, a configuration is described in which one horizontal shaft **126** is provided, but three or more horizontal shafts may be integrally installed with the base unit **23** (**123**).

In the examples and modifications described above, a configuration is described in which the base unit **23** is configured to be rotatable only in the front-and-rear direction, but the base unit **23** may be configured to be rotatable only in the right-and-left direction.

In the above examples and modifications described above, a configuration is described in which the traveling part **2** includes a front traveling body **2**A and a rear traveling body **2**B that are rotatable with each other, but the traveling part **2** may be formed from a single traveling body.

What is claimed is:

**1**. An overhead conveying vehicle in which a main body part having a main frame and a lifting part provided at a lower part of the main frame and configured to elevate a holding part holding a conveyed object is suspended and supported by a traveling part, the overhead conveying vehicle comprising:

a base unit suspended by the traveling part;

a horizontal shaft integrally provided with the base unit and extending in a horizontal direction; and

a pair of support parts provided on the main frame and supporting the horizontal shaft via rubber bushes, wherein

the main body part is suspended and supported by the traveling part via the base unit that is disposed between the traveling part and the main frame in a vertical direction, and the pair of support parts that are provided on a top surface of the main frame.

**2**. The overhead conveying vehicle according to claim **1**, wherein the horizontal shaft is disposed to extend in a direction orthogonal to both a traveling direction of the traveling part and the vertical direction.

**3**. The overhead conveying vehicle according to claim **1**, wherein

a plurality of the horizontal shafts are arrayed along a direction orthogonal to both an extending direction of the horizontal shafts and the vertical direction, and

the horizontal shafts are supported in an unrotatable manner with respect to the base unit and the pair of support parts.

**4**. The overhead conveying vehicle according to claim **1**, wherein

only one horizontal shaft is provided, and

the overhead conveying vehicle further include a pair of dampers provided in the base unit in a manner of sandwiching the horizontal shaft in the traveling direction of the traveling part, and are configured to contact the top surface of the main frame.

**5**. The overhead conveying vehicle according to claim **1**, wherein

the base unit is suspended by the traveling part via a pair of suspension members extending downward in the vertical direction from the traveling part, and

the base unit rotatably supports each of the suspension members around an axis as a turning axis that extends in an extending direction of the suspension members.

**6**. The overhead conveying vehicle according to claim **1**, wherein the rubber bushes respectively provided in the pair of support parts are formed by materials of different hardness from each other.

**7**. The overhead conveying vehicle according to claim **2**, wherein

a plurality of the horizontal shafts are arrayed along a direction orthogonal to both an extending direction of the horizontal shafts and the vertical direction, and

the horizontal shafts are supported in an unrotatable manner with respect to the base unit and the pair of support parts.

**8**. The overhead conveying vehicle according to claim **2**, wherein

only one horizontal shaft is provided, and

the overhead conveying vehicle further include a pair of dampers provided in the base unit in a manner of sandwiching the horizontal shaft in the traveling direction of the traveling part, and are configured to contact the top surface of the main frame.

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